

Table 1. Summary of the Methodology Used to Set the Repulsive Parameters between Beads for Charged Surfactants in Ionic Solutions^a

repulsive parameter	origin	equation
water/water	water compressibility	$a_{ww} = \frac{k_B T}{2\alpha\rho} (\kappa^{-1} N_m - 1)$
like/like	same as water	$a_{ii} = a_{ww}$
tail/water	fitted with the CMC using osmotic pressure	-
charged/water	DFT	$a_{C-W} = 0.132720\Delta G_{hyd} + 123.148$
cationic/anionic	DFT	$a_{ij} = c_1\Delta\Delta G_{hyd} + c_0$
charged/nonionic	DFT	$\Delta a_{C-N} = a_{C-N} - a_{ww} = b\chi \approx b\Delta E/RT$

^a ΔE is the interaction energy between two beads approximated by DFT calculations. b , c_0 , and c_1 are constant numbers. ΔG_{hyd} is the solvation free energy of a charged bead. $\Delta\Delta G_{hyd}$ is the difference of solvation free energy between two charged beads. χ is the Flory–Huggins interaction parameter.

Subsequently, we optimized the repulsion between the tail beads and the water beads, as this interaction plays a crucial role in driving hydrophobicity and significantly affects the CMC of the surfactant.^{12,29,36,55} Sections S3 and S4 in the Supporting Information provide comprehensive lists of all the parameters used. The parameters provided in this work are surfactant independent, and they are consistent for all molecules sharing the same functional group. For instance, any molecule containing an alkyl group with three carbons (a C3 bead) or a carboxylic group (RCOO^-) will use the same set of interaction parameters in the system. Except for the tail–water interaction, the repulsive parameters for the uncharged–uncharged beads and like beads were set equal to the water–water repulsion. Although DFT calculations of these interactions could have been performed to refine the model, their overall contribution to the CMC was considered to be negligible; therefore, the default value for water was used. In contrast, charged beads exhibit an additional interaction force that requires adjustment of their repulsive coefficients with water and other charged beads. To address this, we employed a linear relationship (eq 1) proposed by Nieto-Draghi et al.⁴³ between repulsive parameters (a_{ij}) and the solvation free energy (ΔG_{hyd} , $\text{kJ}\cdot\text{mol}^{-1}$) of ionic species.⁴³

$$a_{\text{ion-water}} = 0.132720\Delta G_{hyd} + 123.148 \quad (1)$$

In their study, Nieto-Draghi et al.⁴³ employed measured individual ion activities⁵⁶ to construct eq 1. Subsequently, these activities were converted into partial osmotic pressures in order to parametrize ion–water interactions. However, as recently emphasized by Warren,⁵⁷ the precise thermodynamic meaning of osmotic pressure in a system is only derived from the combined contributions of both counterions (the cation and the anion) across the membrane (refer to the methodological description below).⁵⁷ Nevertheless, the optimization procedure for eq 1 requires the input of initial guess values for both anions and cations. In the absence of alternative information for initializing the optimization of ion–water interactions, Nieto-Draghi et al.⁴³ proposed that partial osmotic pressures could be used as initial values. This assumption yielded reliable repulsion parameters, as demonstrated below with the variation of CMC with added salinity. Consequently, the aforementioned approach was retained again in our parametrization strategy.

This relationship enabled us to determine the repulsive parameters for the charged beads and water beads. For oppositely charged beads, we used another linear relationship with the difference of solvation free energy ($\Delta\Delta G_{hyd}$), as described by Nieto-Draghi et al.⁴³ and given in eq 2:

$$a_{ij} = c_1\Delta\Delta G_{hyd} + c_0 \quad (2)$$

with c_0 and c_1 being constants of the linear regression for a given cation interacting with anions.

The repulsive parameters for charged–uncharged beads were derived from DFT calculations, further described in Section S3.3 of the Supporting Information.

DPD Methods for CMC Determination. Surfactant molecules self-assemble above the CMC, significantly altering the physical

properties of the solution (Figure 1). Thermodynamically, the CMC manifests as a distinct change in the slope of the osmotic pressure or chemical potential with increasing surfactant concentration. These properties exhibit concentration profiles with a sharp break at the CMC.⁵⁸ Additionally, the phase model of micelles allows for CMC determination from the change in surfactant chemical potential.⁵⁹ Monte Carlo methods permit the calculation of the chemical potential in DPD simulations. However, this method presents large fluctuations due to the soft potentials inherent in DPD and the presence of charged species. This approach is therefore unsuitable for the purposes of this study.

Cluster analyses of concentrated surfactant solutions can also be performed using DPD simulations.^{17–19} Above the CMC, the free surfactant concentration is expected to remain constant at the CMC value. However, DPD simulations present a challenge in estimating the free surfactant concentration due to crowding effects. The phenomenon of crowding is observed when the total surfactant concentration is increased beyond the CMC, resulting in a decrease in the concentration of free surfactant.^{60–65} This behavior is unrealistic and thus requires a mathematical correction.²⁴

Osmotic Pressure Method Using Virtual Walls. Mimicking osmotic pressure experiments in simulations (DPD or atomistic) involves virtual walls interacting with solutes.^{43,66,67} These walls confine solutes by using a recalling harmonic potential. We specifically implemented a new LAMMPS fix to create this recalling force called “wall/harmonic/returned” and a new pair style to accelerate the charged DPD calculations on GPU named “dpd/coul/slater/long”.⁶⁸ The recalling force eliminates ambiguity regarding the control volume and failures on particle–wall overlap in comparison with the already implemented repulsive force. Our approach allows for a soft force while maintaining a defined control volume, independent of the harmonic force strength. Depending on the statistical ensemble, the corresponding framework of the obtained osmotic pressures may vary. As mentioned by Nieto-Draghi et al.,⁴³ the statistical ensemble

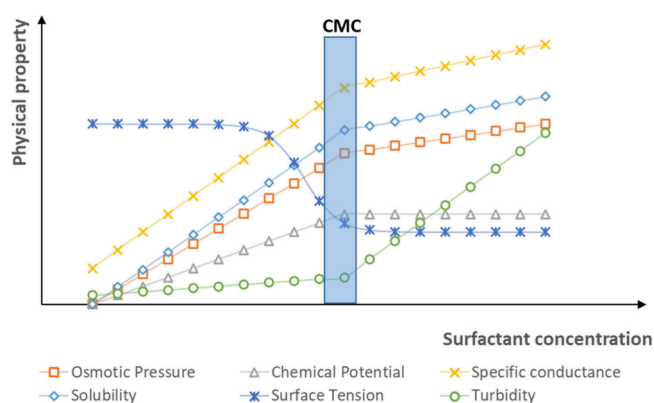


Figure 1. Examples of ideal CMC breakpoints by using different physical properties.